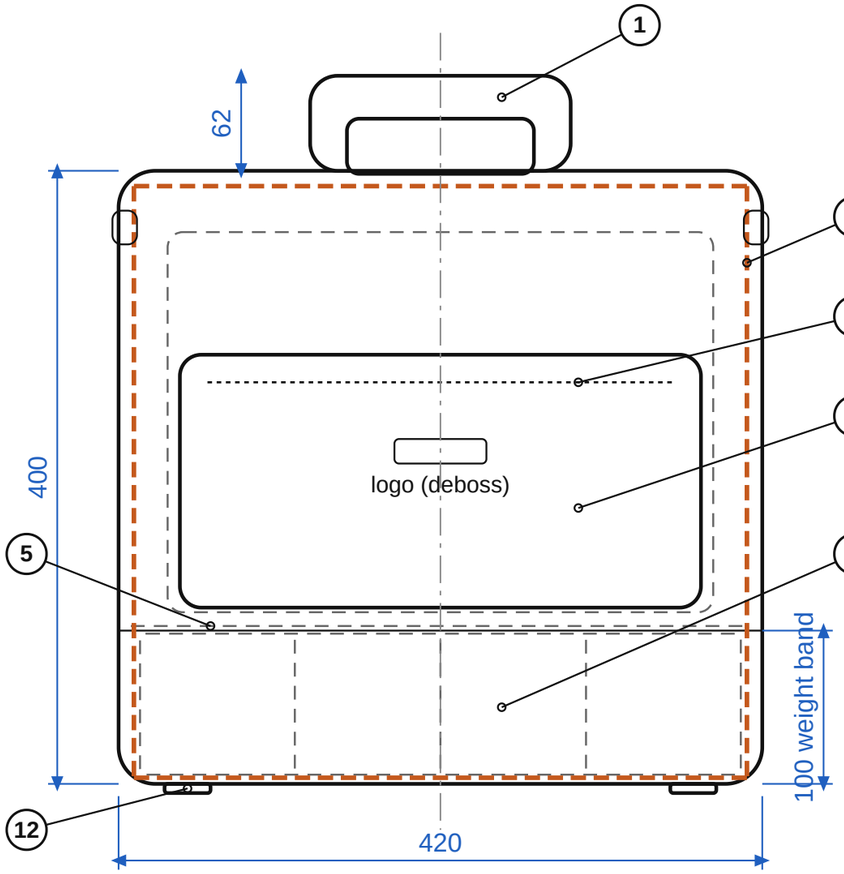
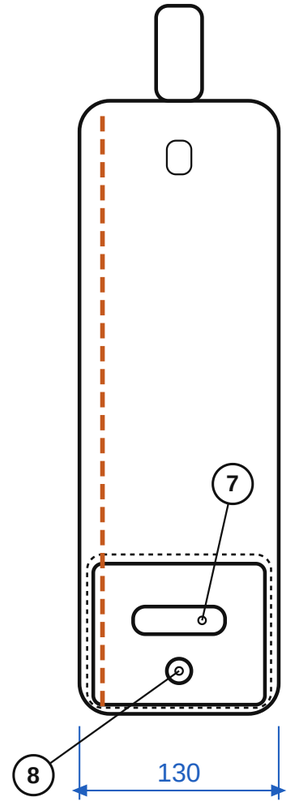
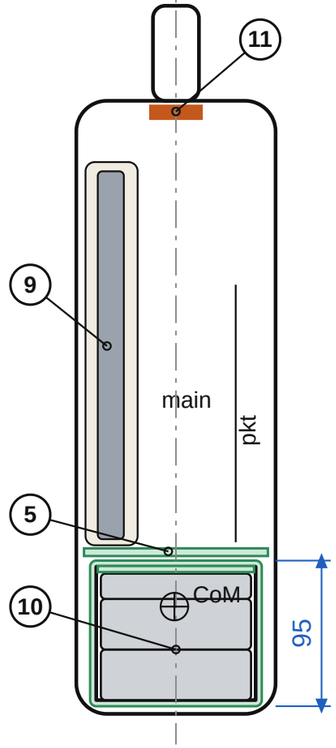


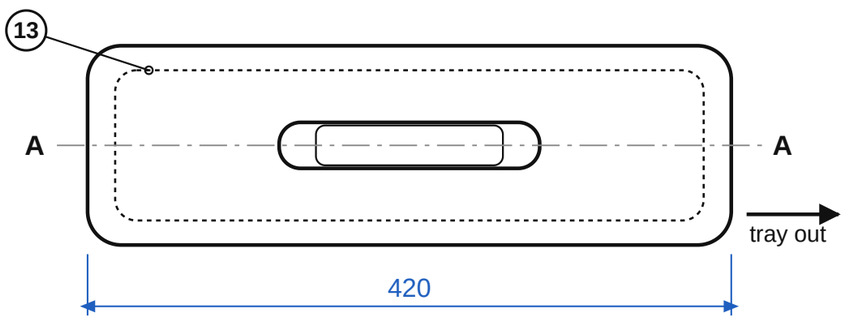
FRONT VIEW



END VIEW



SECTION A–A



TOP VIEW

LAPTOP BAG — “HEAVY BRIEF” (working name)

One-hand suitcase carry, up to 30 kg total (bag + 16" laptop + contents + up to 24 kg weights). Must hang straight and feel the same in either hand.

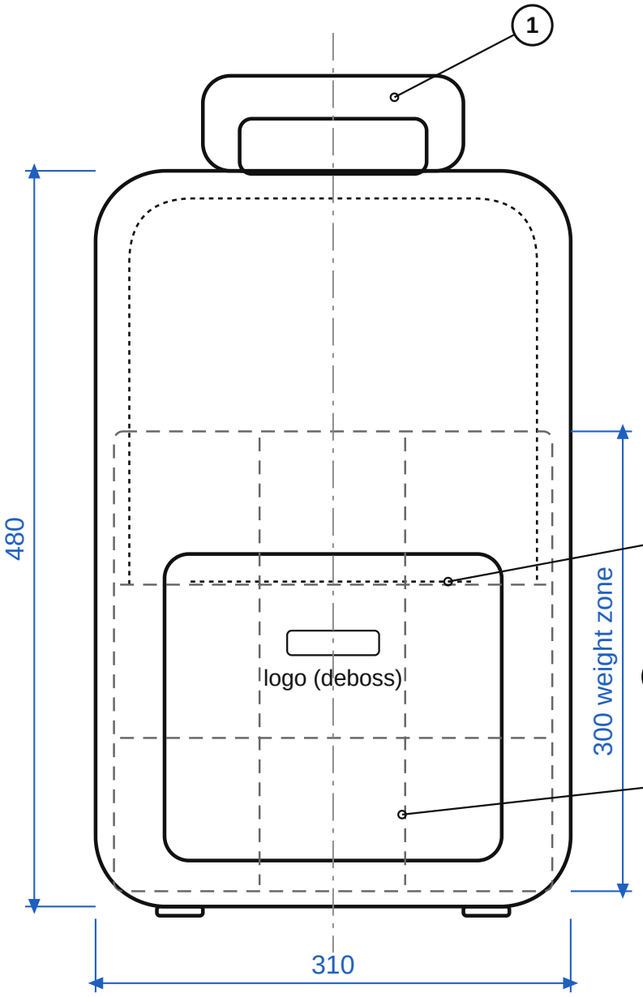
1. Rigid top handle, ~30 mm grip, bolted/riveted to the aluminium bridge (11). Not sewn-only. Rated 90 kg static.
2. Aluminium U-frame: top bridge + one stay in each end panel + bottom cradle under the tray. All load goes handle → frame → tray.
3. 16" MacBook Pro (356 × 248 × 17 mm) in padded sleeve against back wall (9).
4. Weight tray in a 100 mm base band, centred under the handle. 4 modules of 95 × 100 mm; full 24 kg with steel bricks or steel bar. See sheet 3.
5. Rigid HDPE false floor separates the weights from laptop and contents.
6. Front organiser pocket, zipped.
7. Tray slides out of a zipped hatch in one end panel, on HDPE runners. Tray end cap has a pull grip.
8. Two-step release: open hatch zip, then press latch and pull. Tray can't fall out by accident.
10. Clamp plate + 2 cam straps press any stack (10–85 mm) to stop rattle.
12. Four protective base feet.
13. Main zip, 2-way, heavy coil, runs over the top.

Detachable shoulder strap clips to the two D-rings.

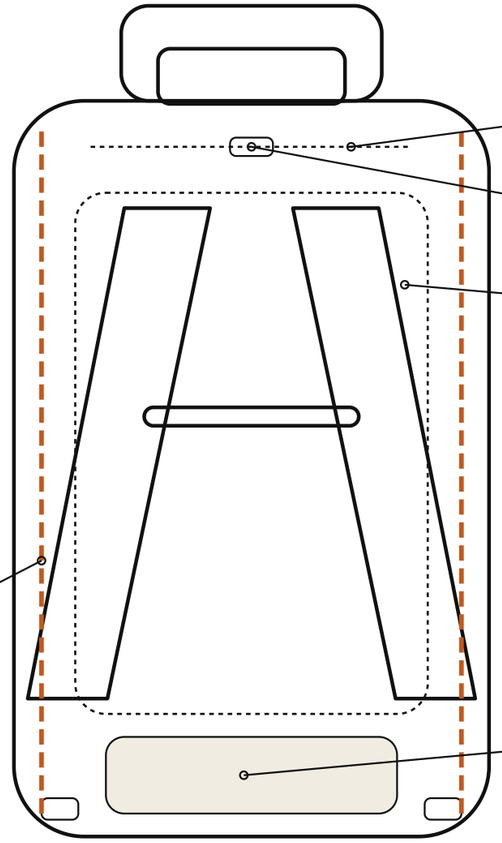
LEGEND

- Visible outline
- Hidden / internal part
- Zip line
- Aluminium frame (6061-T6)
- HDPE sheet / moulded plastic
- Weights
- Centre line / load line
- Loaded centre of mass (CoM)

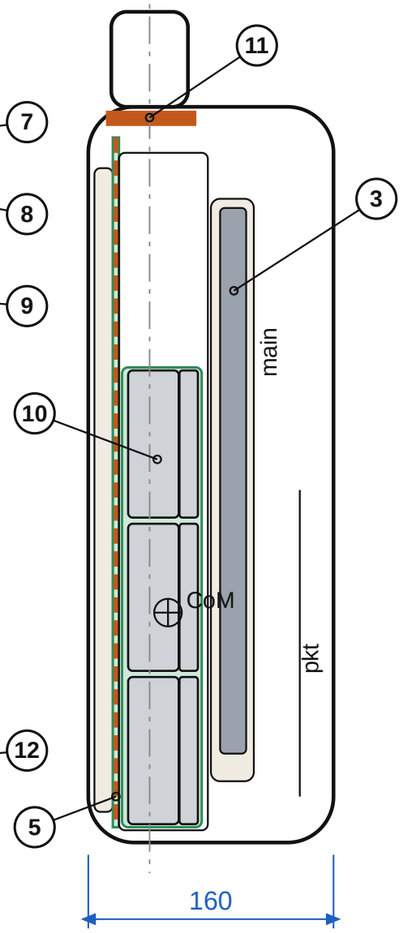
HEAVY BAG — LAPTOP BAG	
Front, end, section and top views Concept sketch · dims in mm · confirm with sample	Sheet 1 of 3 Scale 1:4 at A3



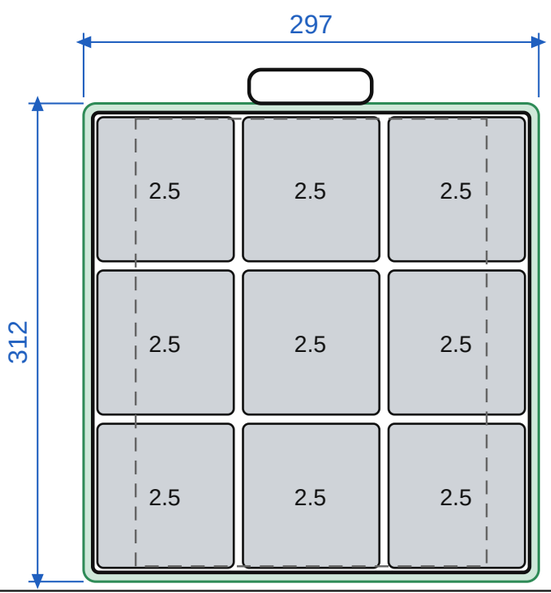
FRONT VIEW



BACK VIEW (straps out)



SECTION B–B



Weight carrier (front, 1:4)

3 × 3 modules of 95 × 100 mm (sheet 3).
Up to 9 steel bricks × 2.5 kg = 22.5 kg,
plus a half-brick layer to reach 24 kg.
Dashed: standard ruck plate 229 × 292 mm
fits the same sleeve (carrier removed).
Lifts out by the pull loop through slot (7).

BACKPACK — “BIG PACK”

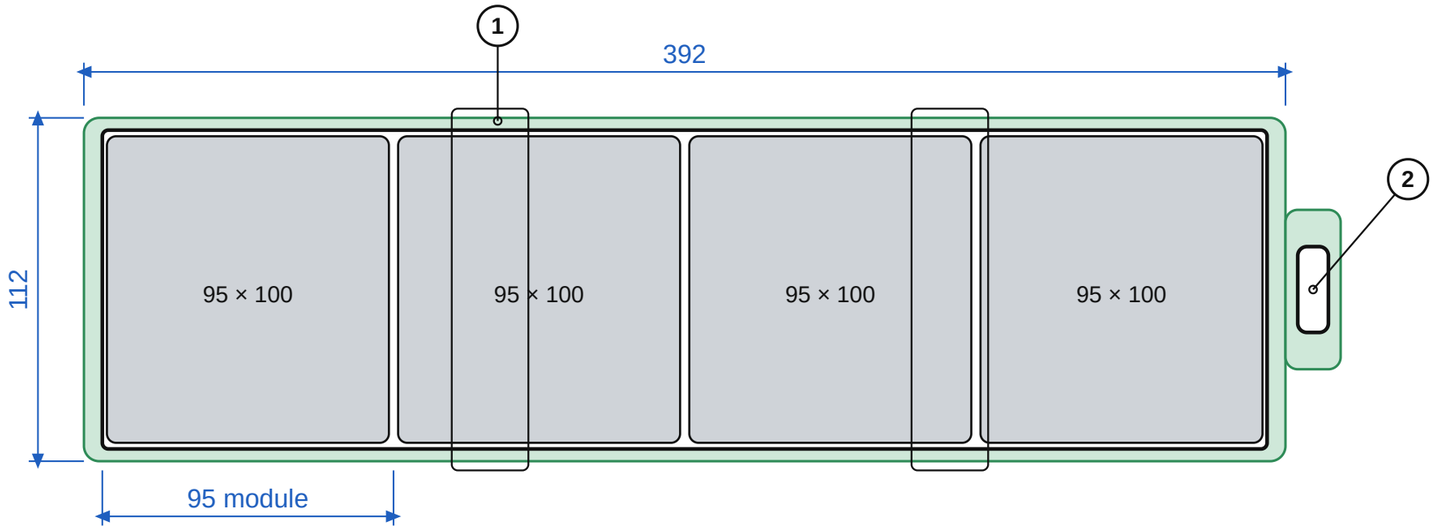
Same 30 kg total rating and one-hand suitcase carry as the laptop bag. Worn as a backpack it's a rest mode (or rucking).

1. Rigid top handle on the aluminium bridge (11), placed over the loaded centre of mass — close to the back panel, not mid-bag.
2. Two aluminium stays in the back panel, joined by the top bridge and a bottom cradle; HDPE frame sheet between them.
3. 16" MacBook Pro in padded sleeve, in front of the weight sleeve.
4. Weight carrier sits low in a sleeve against the frame sheet: weight closest to the spine.
5. Frame sheet and stays carry the load to the bottom cradle.
6. Front organiser pocket.
7. Top-loading weight slot: zip across the top of the back panel; carrier lifts out by its pull loop.
8. Two-step release: buckle tab over the zip pull, then unzip.
9. Shoulder straps tuck behind a zip panel during the hand carry — nothing dangles when switching hands.
10. Carrier holds 3 × 3 bricks, ruck plates, fill-your-own shells or flat plates; cam strap clamps it.
12. Strap anchors, lumbar pad and base feet.

LEGEND

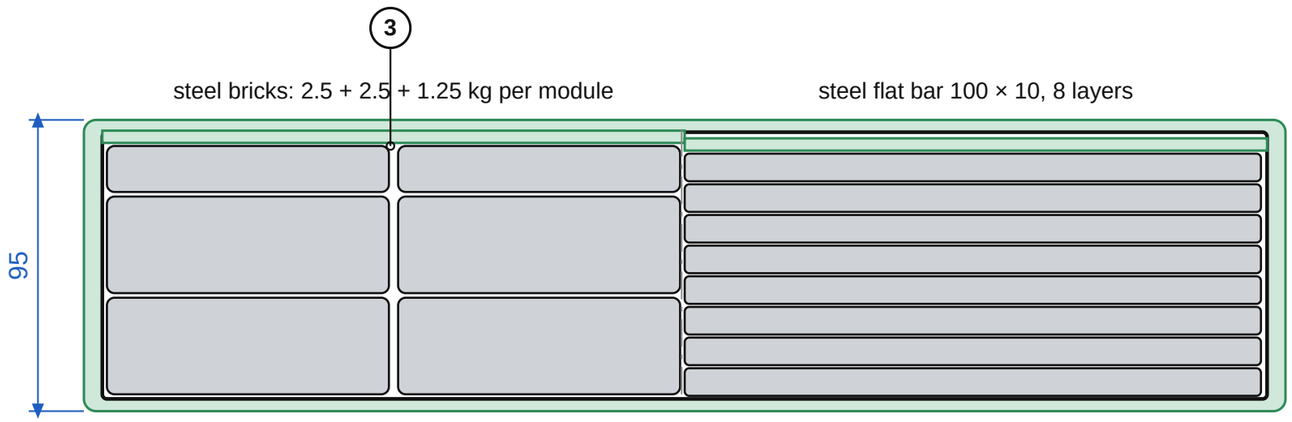
- Visible outline
- Hidden / internal part
- Zip line
- Aluminium frame (6061-T6)
- HDPE sheet / moulded plastic
- Weights
- Centre line / load line
- Loaded centre of mass (CoM)

HEAVY BAG — BACKPACK	
Front, back and section views; weight carrier Concept sketch · dims in mm · confirm with sample	Sheet 2 of 3 Scale 1:4 at A3



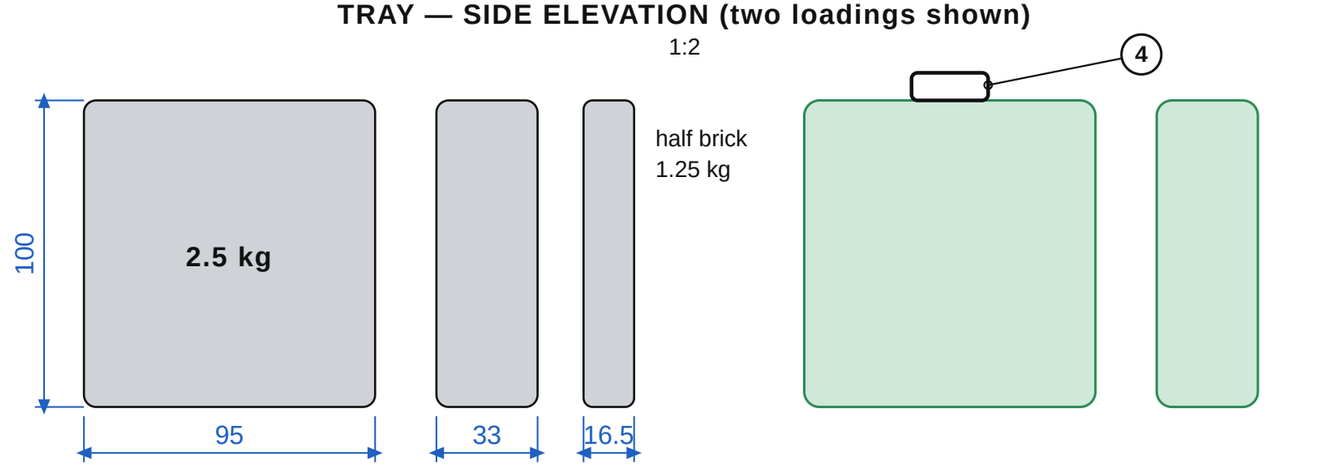
LAPTOP-BAG WEIGHT TRAY — TOP (lid off)

1:2



TRAY — SIDE ELEVATION (two loadings shown)

1:2



STEEL BRICK — front + edge

1:2

FILL-YOUR-OWN SHELL — same footprint

1:2

WEIGHT SYSTEM — “ONE TRAY, ANY WEIGHT”

The bags sell a standard weight interface. Every weight source fits the same module, so one tray (laptop bag) and one carrier (Big Pack) take them all.

MODULE: 95 × 100 mm footprint. Laptop tray = 4 × 1 modules, 85 mm usable height. Big Pack carrier = 3 × 3 modules, 50 mm.

1. Two cam straps press a clamp plate onto any stack from 10 to 85 mm: no rattle with any source.
2. Tray end cap with pull grip and latch; slides on HDPE runners. Tray: aluminium or glass-filled nylon, HDPE liner.
3. Clamp plate (HDPE, 4 mm).
4. Fill-your-own shell: HDPE, screw cap with snap seal. Ships empty; fill with steel shot (~1 kg), sand or water. Long 4-module shell (380 × 100 × 35) for the laptop tray holds ~5 kg of shot.

STEEL BRICK: 95 × 100 × 33 mm steel, 2.5 kg; half brick 16.5 mm, 1.25 kg. TPU or rubber coated, R3 edges, weight laser-marked on face. No lead.

LOCAL STEEL BAR: 100 × 10 mm mild steel cut to 380 mm (3.0 kg, laptop tray) or 285 mm (2.2 kg, Big Pack). We supply TPU sleeves with end caps.

PLATES: standard ruck plate 229 × 292 mm goes straight into the Big Pack sleeve. 1" gym plates up to 150 mm diameter lie flat in the Big Pack carrier. Gym plates don't fit the laptop tray.

CAPACITY: laptop tray 24 kg with bricks or bar; ~13–16 kg with shot shells. Big Pack 24 kg with any source.

LEGEND

- Visible outline
- Hidden / internal part
- Zip line
- Aluminium frame (6061-T6)
- HDPE sheet / moulded plastic
- Weights
- Centre line / load line
- Loaded centre of mass (CoM)

HEAVY BAG — WEIGHT SYSTEM	
Tray, carrier module, bricks and shells Concept sketch · dims in mm · confirm with sample	Sheet 3 of 3 Scale 1:2 at A3